A Study on the Impact of Screen Curvature and Control-Display Gain on Pointing Performance in Large Displays

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Abstract. This study investigates the impact of screen curvature and control-display (C-D) gain on pointing performance in large displays. Thirty participants performed Fitts' tasks on flat, 1500R, and 800R displays with small and large C-D gains. Results indicated that large C-D gain significantly reduced movement time, while the effect of curvature was partially observed. Throughput was also higher with large C-D gain, suggesting its effectiveness in large displays. The findings underscore the importance of considering appropriate C-D gain according to form factors of display to optimize pointing performance. This research highlights the need for higher C-D gains for better performance and possible performance risks associated with display curvature in large displays. These findings provide a basis for future studies on optimizing pointing performance for large displays, potentially informing the design of advanced interfaces for various applications.

Keywords: Fitts' Law, Control-Display gain, Large Display, Curved Display, Pointing Performance.

1 Introduction

The advent of large displays is reshaping the landscape of consumer electronics, blurring the lines between traditional TVs and monitors. The growing consumer desire for enhanced immersion and productivity has driven a significant shift towards larger screens across various contexts such as gaming, video watching, and working environments [1], [4]. With the growth of large displays, curved displays have emerged as a solution to some critical drawbacks of large flat displays, such as visual fatigue and distracting reflections [11]. Previous study on curved display has reported beneficial effects of curved compared with flat in large displays [5], possibly attributed to wider field-of-view (FOV) by providing equal viewing distances across the screen.

Control-Display (C-D) gain becomes particularly significant when interacting with these large, curved displays. The C-D gain is defined as the ratio of the distance moved on the display to the distance moved by the control device, affecting how users perceive and control their interactions with the screen. Previous studies have proposed an optimal C-D gain for young users as 4, based on experiments using a 17" CRT screen [8]. However, these findings may not be directly applicable to larger displays. Therefore,

our study focuses on investigating how curvature, C-D gain, and their interaction impact on large displays.

To investigate these factors, Fitts' Law provides a robust framework. Fitts' Law, widely used to evaluate human motor system performance in HCI, establishes a relationship between movement time (MT) and the variables of movement amplitude (A) and target width (W) [3]. Throughput (TP), also known as the index of performance, is a measure that combines both speed and accuracy of movement performance. The units of throughput are bits per second(bits/s). The Shannon's formulation index of difficulty (ID), using the unit bits, was adjusted to the effective index of difficulty (IDe) in this study [2], as IDe captures the actual movement of a participant rather than the experimenter's intention. IDe can be calculated by adjusting the target width to effective target width that includes spatial variability or accuracy in responses [7]:

$$IDe = log_2(A/We + 1) \tag{1}$$

$$We = 4.133 \cdot SD_x \tag{2}$$

The primary objective of this research is to determine whether existing interface designs are adequate for future large displays or if new standards are necessary. By examining the effects of curvature and C-D gain on pointing performance, this study aims to provide insights into the optimal conditions for interface design in large displays.

2 Method

2.1 Participants

30 participants (10 females, 20 males) were recruited for the experiment. The average age of the participants was 26.2, and standard deviation was 4.11. All participants were right-handed, competent computer users, and 27% had a large monitor experience which were larger than 40 inches.

2.2 Experiment Design

We used 3 * 2 within-subjects design with repeated measures. The independent variables were three levels of display curvature (Flat, 1500R, 800R) and two levels of C-D gain values (small (4) and large (12.7)). C-D gain was set by adjusting mouse speeds to 4 and 6 to achieve C-D gain values of 4 and 12.7 respectively. The dependent variables were Movement Time (MT) and Throughput (TP) calculated based on the Fitts' Law model formula [6]. Intercept a and slope b are empirically determined constants.

$$MT = a + b \cdot ID \tag{3}$$

Two calculation approaches were analyzed to investigate throughput: mean-of-means (TP_{avg}) [9] and slope-inverse (TP_{inv}) throughput [12]. Either way, a grand throughput was produced for each condition. The formulas for each throughput calculation are:

$$TPavg = \frac{1}{y}\sum_{i=1}^{y}(\frac{1}{x}\sum_{j=1}^{x}\frac{IDe_{ij}}{MT_{ij}}) \quad (4)$$

$$TPinv = 1/b \quad (5)$$

Three large displays representing each curvature condition were prototyped for the lab experiment. To enable participants to perform tasks on the display, a projector behind the display projected images onto the rear screen (Figure 1). Length of the large displays was 145 cm with a screen resolution of 1366 * 768 (px).

Fig. 1. Experiment environment using 800R display mockup

2.3 Procedure

Participants were first provided with an explanation of the study and asked to sign a voluntary consent form. Before the start of the experiment, they were instructed to perform the Fitts task as quickly and accurately as possible. The experiment consisted of 12 trials, with the first 2 trials serving as practice to familiarize participants with each condition. To minimize order effects, the sequence of conditions was counterbalanced for each participant. Verbal questions were also asked post-experiment to collect subjective perceptions on the overall usability of mouse control in relation to curvature and mouse speed, specifically whether participants had a negative, positive attitude, or neutral attitude toward these variables.

2.4 Data Analysis

The Fitts' Law task was demonstrated by the FittsStudy program [10], setting target widths of 32, 64, 128 (px) and amplitudes of 256, 640, 1024 (px) which follow 3.4, 6.8, 13.6 (cm) and 27.2, 67.9, 108.7 (cm), respectively. Therefore, a total of 8 Index of Difficulty (ID) values were repeated for 12 trials per condition. Although 648 data points were collected per participant, the first two practice trials (108 data points) were excluded, resulting in a total of 16,200 data points used for analysis by repeated measures ANOVA.

3 Results

3.1 Movement Time and Throughput

Repeated measures ANOVA analysis found significantly shorter movement time with large C-D gain condition compared with small C-D gain condition ($Mean\ MT_{large}$: 0.946, $Mean\ MT_{small}$: 1.015, F=122.252, p<0.001). Overall, the effect of curvature on movement time was not significant. However, when the C-D gain was large, the flat condition resulted in a shorter movement time ($Mean\ MT_{flat}$: 0.964, $Mean\ MT_{1500R}$: 0.990, $Mean\ MT_{800R}$: 0.986, F=7.331, p=0.001).
For throughput, there were no significant differences between curvature and C-D gain in TP_{avg}. However, a significantly higher TP_{inv} was observed at large C-D gain compared with small condition (F=28.364, p<0.001). Table 1 outlines both TP_{avg} and TP_{inv} data mean and standard deviation for each condition.

			Control-Display Gain									
			4					12.7				
			a	*b*	*r*	*TPavg*	*TPinv*	*a*	*b*	*r*	*Tpavg*	*TPinv*
Curvature	Flat	Mean	27.73	258.99	0.86	3.92	4.12	196.88	204.69	0.85	3.97	5.14
		SD	201.27	69.95	0.10	0.65	1.03	138.47	47.78	0.08	0.63	1.14
	1500R	Mean	10.02	264.80	0.86	3.93	4.04	161.94	225.77	0.85	3.89	4.77
		SD	218.11	69.94	0.08	0.70	1.02	148.89	60.38	0.06	0.83	1.33
	800R	Mean	-16.17	275.34	0.85	3.89	3.98	160.85	217.90	0.84	4.00	4.96
		SD	256.59	79.79	0.10	0.74	1.26	203.02	65.67	0.09	0.73	1.35

Table 1. Summary of Fitts' Law results with regression coefficients (intercept *a* and slope *b*), correlation coefficients *r*, mean-of-means(TP_{avg}) and slope-inverse(TP_{inv}) throughput.

3.2 Modeling Movement Time with ID

As shown in Figure 3, it was observed that the lower index of difficulty (ID), which corresponds to shorter distances and wider bandwidths, exhibits negligible influence from the C-D gain. Conversely, as the ID value increases, the impact of C-D gain becomes more prominent, with smaller C-D gains resulting in longer movement time (MT). Although the differences were not statistically significant, there was a noticeable trend indicating that the flat display condition yielded lower MT compared to the curved display condition. Figure 2 illustrates that, despite no significant differences in MT was found under the 800R condition, the largest discrepancy is observed at higher index of difficulty (ID) levels compared with the flat or 1500R conditions.

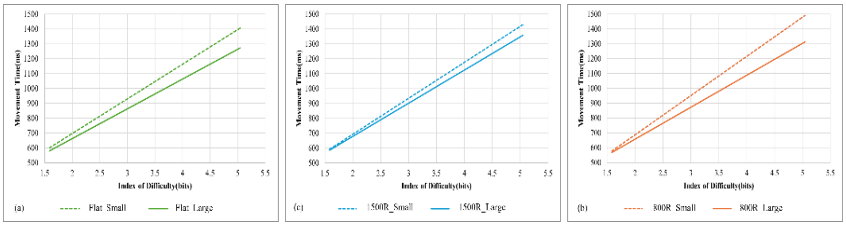

Fig. 2. Best-fit lines of the scatter plot to compare the effect of C-D gain in each curvature. (a), (b), (c) represent flat, 1500R, 800R condition respectively.

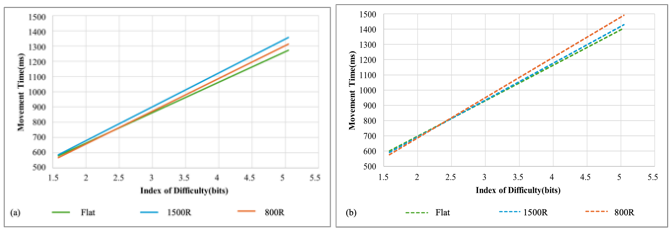

Fig. 3. Best-fit lines of the scatter plot to compare the effect of curvature in each C-D gain. (a) and (b) represent large and small C-D gain condition respectively.

3.3 Subjective Evaluation

After completing all tasks, participants were briefly interviewed about any difficulties or preferences related to mouse speed and curvature. Meaningful excerpts from their responses are presented. One participant mentioned, "When there is curvature, the speed seems to move faster than I intended at the curved parts, so it was most difficult to control when the curvature was severe." Another participant expressed a preference for the flat display, stating, "It's more comfortable because I'm more accustomed to it." Regarding speed, some participants found it easier to control the mouse at higher speed (large C-D gain), while others found it inconvenient to move the mouse multiple times at slower speed (small C-D gain). Interestingly, several participants noted that turning their necks was easier with curved displays, while they found it visually inconvenient with the flat display when the edges of the screen were not visible at a glance.

4 Discussion

The results confirmed the main effect of C-D gain in movement time (MT). In terms of throughput (TP), a significantly higher throughput was observed in TP_{inv} with large C-D gain, suggesting large rather than small C-D gain might be more effective in large displays considering pointing performance. One possible explanation for the difference in pointing performance is the difficulty in controlling movements on curved displays due to equidistant viewing from all angles. This suggests that as screen curvature increases, a distortion effect may lead participants to underestimate the distance they have traveled. Consequently, future research should also examine error rates alongside movement time to determine if the increased movement time observed with the 800R display can be attributed to participants overshooting the target and subsequently having to correct their course. Indeed, subjective evaluation indicated difficulties in adjusting mouse speed at the edges of the display as the curvature increased.

In the sense of subjective evaluation, despite throughput showing significant differences only in terms of speed, comments about curvature were more frequent than those about speed. This might imply that curved displays might have more impact on subjective user experience, such as individual preferences and comfort, while throughput was higher with flat displays.

5 Conclusion

This study explored the effects of curvature and C-D gain on the performance of Fitts task in large displays. Significant effects of C-D gain and the presence of curvature in large C-D gain condition on performance were revealed. The result suggests that higher C-D gain may be preferable for large displays, and curvature influences control performance. These findings are expected to serve as foundational data for future research on the optimal pointing performance in large displays. As large displays are anticipated to expand their applications beyond traditional uses, such as for smart factory manipulation or entertainment in smart cities, the present study is expected to be applied and utilized in various future studies.

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